

Supplementary Materials

1 Proof of Theorems

In this section, we provide the proof of Theorems 1, 3, and Proposition 2.

1.1 Molecular Explanation

We begin by proving that ESPAM satisfies the three axioms of Efficiency, Symmetry, and GNN-aware fair treatment as indicated in Theorem 1.

Theorem 1. • Efficiency:

$$\begin{aligned}
 \sum_{i \in V} \phi_i(g) &= \sum_{i \in V} \frac{1}{L} \sum_{\ell=1}^L \phi_i^{(\ell)}(g) = \\
 \frac{1}{L} \sum_{i \in V} \sum_{\ell=1}^L \phi_i^{(\ell)}(g) &= \frac{1}{L} \sum_{\ell=1}^L \sum_{i \in V} \phi_i^{(\ell)}(g) \stackrel{\text{Efficiency of FESP}}{=} \\
 \frac{1}{L} \sum_{\ell=1}^L g(V) &= g(V)
 \end{aligned}$$

- Symmetry: As each game of ego-networks follows this axiom, their average also follows the same axiom.
- GNN-aware fair treatment:

$$\begin{aligned}
 \forall \ell : \begin{cases} g(Ego_\ell(v)) \geq g(Ego_\ell(u)), \\ g(N \setminus Ego_\ell(v)) \leq g(N \setminus Ego_\ell(u)) \end{cases} &\Rightarrow \\
 \forall \ell : \begin{cases} w^{(\ell)} g(Ego_\ell(v)) \geq w^{(\ell)} g(Ego_\ell(u)), \\ (1 - w^{(\ell)}) g(N \setminus Ego_\ell(v)) \leq \\ (1 - w^{(\ell)}) g(N \setminus Ego_\ell(u)) \end{cases} &\Rightarrow \\
 \forall \ell : w^{(\ell)} g(Ego_\ell(v)) - (1 - w^{(\ell)}) g(N \setminus Ego_\ell(v)) \geq \\
 w^{(\ell)} g(Ego_\ell(u)) - (1 - w^{(\ell)}) g(N \setminus Ego_\ell(u)) &\Rightarrow \\
 \forall \ell \phi_v^{(\ell)}(g) \geq \phi_u^{(\ell)}(g) \Rightarrow \phi_v(g) \geq \phi_u(g)
 \end{aligned}$$

□

1.2 Interaction Explanation

First, we prove Proposition 2, which indicates the FESP values in Equation 7 satisfy the Efficiency axiom.

Proposition 2. As the game of each row is an FESP game, it follows the efficiency axiom. Therefore, the sum of the columns in a row is equal to $\text{sigmoid}(\phi_i(g) - \phi_i^N(g))$. It should be noted that when all the segments are masked, all the model predictions become zero, resulting in the ESPAM value of each node being zero. Therefore, $\phi_i^N(g)$ becomes zero too. Hence, $\text{sigmoid}(\phi_i(g) - \phi_i^N(g)) = \text{sigmoid}(\phi_i(g))$. $\square$

In the next step, we provide the proof of Theorem 3, which provides our desired theoretical guarantees for explanations of receptor-molecule interactions.

Theorem 3. • Efficiency: By the definition of g in the theorem and Proposition 2, it is deducible.

- Symmetry: As each term of $\sum_{i \in \text{atoms}} C_{ij}$ comes from an FESP game which respects the symmetry axiom, the sum itself also respects the symmetry axiom.

- Fair Treatment

$$\begin{aligned}
& \forall S : g(S \cup \{i\}) \leq g(S \cup \{j\}) \Rightarrow \\
& \forall S : \sum_{k \in \text{atoms}} \text{sigmoid}(\phi_k - \phi_k^{S \cup \{i\}}) \leq \\
& \sum_{k \in \text{atoms}} \text{sigmoid}(\phi_k - \phi_k^{S \cup \{j\}}) \Rightarrow \\
& \left\{ \begin{array}{l} \sum_{k \in \text{atoms}} \text{sigmoid}(\phi_k - \phi_k^{\emptyset \cup \{i\}}) \leq \\ \sum_{i \in \text{atoms}} \text{sigmoid}(\phi_i - \phi_i^{\emptyset \cup \{j\}}) \\ \text{and} \\ \sum_{k \in \text{atoms}} \text{sigmoid}(\phi_k - \phi_k^{N \setminus \{i,j\} \cup \{j\}}) \geq \\ \sum_{i \in \text{atoms}} \text{sigmoid}(\phi_i - \phi_i^{N \setminus \{i,j\} \cup \{i\}}) \end{array} \right. \quad 0 \leq w_k \leq 1 \\
& \sum_{k \in \text{atoms}} w_k \text{sigmoid}(\phi_k - \phi_k^{\emptyset \cup \{i\}}) - \\
& \sum_{k \in \text{atoms}} (1 - w_k) \text{sigmoid}(\phi_k - \phi_k^{N \setminus \{i,j\} \cup \{j\}}) \leq \\
& \sum_{k \in \text{atoms}} (1 - w_k) \text{sigmoid}(\phi_k - \phi_k^{\emptyset \cup \{i\}}) - \\
& \sum_{k \in \text{atoms}} (1 - w_k) \text{sigmoid}(\phi_k - \phi_k^{N \setminus \{i,j\} \cup \{j\}}) \Rightarrow \\
& \sum_{k \in \text{atoms}} C_{ki} \leq \sum_{k \in \text{atoms}} C_{kj}
\end{aligned}$$

□